RAJINDER LAW ACADEMY CRM

Master Production Specification & 7-Day Rapid Sprint Developer Roadmap

Exclusively Undergraduate (UG) • Cohort-Specific Sidebars & Content • Strictly One-Time Payments

COHORTS: 1. STATE JUDICIARY (UG) | 2. CLAT UG | 3. PANJAB UNIVERSITY LAW (PU CET / 3-YR LLB)

CLIENT ARCHITECTURAL MANDATES (STRICTLY ENFORCED):

1. ZERO INSTALLMENTS: The concept of installment schedules is completely removed. Only One-Time Full Course Payments are accepted. Every fee record represents a single upfront admission/tuition fee with an instant single GST Tax Invoice.
2. STRICTLY UNDERGRADUATE (UG): No Post-Graduate (PG) content, no CLAT PG, no PU LL.M, and no Higher Judicial Services (HJS).

3. DEDICATED COHORT SIDEBARS & CONTENT: Students in Judiciary, CLAT UG, and Panjab University Law must see entirely distinct sidebar tabs, tailored study material, and relevant test engines. CLAT students never see Mains DAW or Bare Act decoders.

1. The Three Distinct Undergraduate Legal Cohorts

Attribute	Judicial Services (UG Track)	CLAT UG (5-Year Law)	Panjab University Law (PU CET / LLB)
Target Aspirant	UG Law Students (5-Yr/3-Yr) & Fresh Law Graduates targeting State Judiciary.	Class 11, Class 12 & Droppers targeting 26+ National Law Universities (NLUs).	Class 12 students (5-Yr BA/BCom CET) & Non-Law Graduates (PU 3-Yr LLB PGLAW).
Target Examination	Civil Judge Junior Division (DJS, HCS-J, UP PCS-J, RJS, MP-CJ, BPSC-J).	CLAT UG, AILET (NLU Delhi), SLAT (Symbiosis), LSAT India.	PU BA/B.Com LL.B (5-Yr) CET & PU LL.B (3-Yr) Entrance Test, Chandigarh.
Testing Engine	Bare Act Sectional MCQs + Daily Subjective Mains Answer Writing (DAW).	100% Passage-based Comprehension MCQs (120 Qs / 120 Mins / -0.25 negative).	100 Objective MCQs (Legal Aptitude, Current Affairs & GK, Mental Ability, English).
Primary Knowledge Assets	Major Acts (BNS, BNSS, BSA, CPC, Constitution), Case Laws, Latin Maxims.	Fact-Principle Reasoning Vault, Monthly GK Compendiums, Critical Reasoning.	PU Law Syllabus Notes, Constitution Aptitude, PU Solved Past 10 Years Papers.
Pricing & Payment Model	One-Time Course Fee (₹45,000–₹1,50,000) with i GST Tax Receipt.	One-Time Course Fee (₹35,000–₹85,000) with ins Tax Receipt.	One-Time Course Fee (₹25,000–₹55,000) with instant ful Tax Receipt.

2. Dedicated Student Sidebar Navigation Directory (Side-by-Side Matrix)

Exact specification of the three dynamic sidebar menus rendered by StudentSidebar.tsx based on user.academicStream. Every cohort receives a 100% custom menu reflecting their entrance examination format:

1. STATE JUDICIARY (UG)

- Classroom Overview
- Enrolled Judiciary Courses
- Live Classroom Chamber (100dvh)
- Batch Timetable & Shifts
- Daily Legal Digest (Ratio)
- Prelims State-wise Mock Series
- Bare Act Practice Quizzes
- Mains Daily Answer Writing
- Evaluated Copies & Remarks
- Bare Act Comparative Tables
- Landmark Judgments Digest
- Solved Judicial Past Papers
- Digital Student QR ID Pass
- Chamber Doubts & Faculty Chat
- Course Fee Receipt (One-Time)
- Academy Notice Board

2. CLAT UG

- CLAT Classroom Overview
- My CLAT Courses (Found./Crash)
- Live CLAT Interactive Chamber
- CLAT Batch Timetable & Shifts
- Daily Legal & Current GK Brief
- All-India NLU Mocks (120 Qs)
- Sectional Passage Drills
- NLU Cutoff Rank Predictor
- Error Notebook (Mistake Vault)
- Starred Questions Vault

[MAINS WRITING STRICTLY HIDDEN]

- Fact-Principle Reasoning Vault
- Monthly Current Affairs Vault
- Critical Reasoning & Vocab
- CLAT Official Solved PYQs
- Digital Student QR ID Pass
- NLU Mentor Connect & Doubts
- Course Fee Receipt (One-Time)
- Academy Notice Board

3. PANJAB UNIVERSITY LAW

- PU Law Overview & Campus Hub
- PU 5-Yr / 3-Yr Batches
- Live PU Prep Classroom
- PU Batch Timetable & Shifts
- Daily Legal Aptitude & GK Feed
- PU CET / LLB Full Length Mocks
- Legal Aptitude Sectional Drill
- GK & Mental Ability Quizzes
- PU Cutoff Predictor
- Error Notebook (Mistake Vault)
- PU Law Syllabus & High-Yield
- PU 10-Yr Solved Past Papers
- PU Dept of Laws Circulars
- Digital Student QR ID Pass
- PU Faculty Mentorship & Doubts
- Course Fee Receipt (One-Time)
- Academy Notice Board

DEVELOPER CODE IMPLEMENTATION (FRONTEND):

In frontend/src/navigation/student-sidebar-items.ts, export function getSidebarItemsForStream(stream: string). In StudentSidebar.tsx, subscribe to useAuthStore((s) => s.user?.academicStream). The navigation bar automatically re-renders the exact menu corresponding to their cohort with zero code duplication.

3. Essential New Features Added to User & Admin Panels

To transform the academy CRM into a commercial-grade, multi-cohort enterprise platform, the following high-impact features are incorporated across both portals:

Feature Name	Portal	User / Student Experience	Admin Control & Architecture
One-Time Fee Billing & GST Invoicing	**User & Admin**	Student views clean "Paid in Full" or "Payment Pending" card. Instant 1-click download of authentic GST Tax Invoice PDF with RLA seal and HSN code.	Complete removal of installment tracking. Admin records full payment via UPI/Cash/NetBanking, issues receipt, and views total gross revenue metrics.
NLU & PU Cutoff Rank Predictor	**User Portal**	On completing full-length mock tests, student sees dynamic benchmark indicating predicted percentile and likelihood of getting into Tier 1/2/3 NLUs or PU Dept of Laws.	Admin configures historical cutoff marks for top colleges (NLSIU, NALSAR, WBNUJS, PU Chandigarh) under System Settings.
Error Notebook (Mistake Vault)	**User Portal**	Every MCQ answered incorrectly during mock tests or quizzes is automatically saved into a dedicated "Mistake Notebook". Student can re-quiz strictly on past mistakes.	System tracks question error rate across student cohorts to identify universally tricky questions.
Live Class PDF Handouts & Lecture Slides	**User & Admin**	Attached directly underneath the Cinema Lecture Player and Live Classroom Chamber. 1-click download of faculty whiteboard notes and class presentation PDFs.	Admin/Faculty uploads slide handouts (PDF) directly when creating/editing a lecture or launching a live class.
Multi-Cohort Admin Switcher Dropdown	**Admin Portal**	N/A (Admin tool)	Global top-nav dropdown in Admin Command Center: Switch view between "All Cohorts", "Judiciary", "CLAT UG", and "PU Law". Instantly filters KPIs, students, and courses.
Admissions 60-Sec Fast Enrollment	**Admin Portal**	Student receives immediate SMS/WhatsApp notification with roll number and login link.	One-page streamlined enrollment form: captures student details, selects course, records one-time payment, auto-generates roll no RLA-YYYY-STREAM-XXX, and activates account.
Printable Digital QR ID & Gate Pass	**User & Admin**	Student downloads high-res printable ID card PDF with photo, roll number, stream badge, and dynamic QR gate token.	Campus guards/turnstiles use scan API (POST /campus/verify-gate-pass) to verify student admission status and log campus entry timestamps.
Bulk CSV Question Importer CMS	**Admin Portal**	N/A (Admin tool)	Uploads 100+ questions in seconds. Distinct template support for: (1) Standard Law MCQs, (2) CLAT Reading Passages + Child Questions, (3) PU CET Sectional MCQs.

4. Full-Stack Architecture: Models, APIs & Portals

Module	Database Model & Schema	Backend API Routes	Frontend & Admin Key Files
Authentication & User Profiling	UserSchema: • academicStream: ["judiciary", "clat_ug", "punjab_university_ug"] • academicStatus: Class 11, Class 12, Law Student, Law Graduate, Full-Time • feeStatus: "Paid in Full" | "Pending"	• POST /api/v1/auth/login • POST /api/v1/auth/onboarding • POST /api/v1/auth/google • GET /api/v1/auth/me	• frontend/src/pages/auth/OnboardingPage.tsx • frontend/src/components/dashboard/StudentSidebar.tsx • admin/src/pages/students/StudentsDirectoryPage.tsx
One-Time Fee Billing & Invoicing	StudentFeeSchema: • student, course, totalFee: Number • paymentStatus: "Paid in Full" | "Pending" • paidAmount: Number, paidAt: Date • paymentMode: UPI / Cash / NetBanking • invoiceNumber: String, receiptUrl: String	• GET /api/v1/fees/my-status (student full fee) • POST /api/v1/fees/admin/record-payment (marks Paid, issues invoice no) • GET /api/v1/fees/admin/all (stream filter)	• frontend/src/pages/dashboard/FeesPage.tsx • frontend/src/utils/generateInvoicePdf.ts • admin/src/pages/campus/FeesPage.tsx
CLAT & PU Test Engines	MCQQuestionSchema: • stream: judiciary | clat_ug | pu_law • passageText: String (optional, CLAT) • passageRefCode: String • subject, question, options, correctOption MockTestSchema: • durationMinutes, maxScore, negativeWeight	• POST /api/v1/tests/passage-set • GET /api/v1/tests/mock-series?stream=... • POST /api/v1/test-engine/submit • GET /api/v1/test-analytics/my-results	• frontend/src/components/tests/TestExamModal.tsx (Split-Screen for CLAT) • frontend/src/pages/dashboard/tests/TestResultsPage.tsx • admin/src/pages/exams/TestsQuestionBankPage.tsx
Academic Lectures & Live Chamber	CourseSchema: category (Judiciary/CLAT/PU) LiveClassSchema: ugCohort, streamStatus VideoWatchSessionSchema: watchedSeconds	• GET /api/v1/courses?stream=... • GET /api/v1/live-classes/current • POST /api/v1/courses/track-watch	• frontend/src/pages/dashboard/courses/LecturePlayerPage.tsx • frontend/src/pages/dashboard/LiveClassesPage.tsx (100dvh) • admin/src/pages/academic/LiveClassesPage.tsx
Knowledge Vaults & Solved PYQs	PyqPaperSchema: examCategory (Judiciary, CLAT_UG, PU_CET_5YR, PU_LLB_3YR) BareActNoteSchema: oldAct, newAct JudgementDigestSchema: bench, ratioDecidendi	• GET /api/v1/pyqs?stream=... • GET /api/v1/bare-acts • GET /api/v1/judgements • GET /api/v1/daily-digest/today	• frontend/src/pages/dashboard/PyqArchivePage.tsx • frontend/src/pages/dashboard/DailyDigestPage.tsx • admin/src/pages/vault/PyqsPage.tsx
Admissions & Campus Operations	AdmissionLeadSchema: ugStream, stage, source UserSchema: rfidTurnstileId, gateLogs: [{ timestamp, gate, direction }]	• GET /api/v1/leads?stream=... • PATCH /api/v1/leads/:id/stage • POST /api/v1/leads/enroll (fast enrollment) • POST /api/v1/campus/verify-gate-pass	• admin/src/pages/leads/LeadsPipelinePage.tsx • admin/src/pages/leads/LeadsNewEnrollmentPage.tsx • frontend/src/pages/dashboard/StudentIdPage.tsx

5. Accelerated 7-Day Developer Sprint Plan (Day 1 to Day 4)

A battle-tested 7-day engineering roadmap. Each day outlines the exact files to edit, concrete implementation tasks, and measurable acceptance criteria:

Day	Core Focus	Concrete Implementation Tasks & Code Changes	Files Modified	Acceptance Criteria
Day 1	**Multi-Cohort Schema & One-Time Payment Model**	• Refactor User, Course, MockTest, PyqPaper schemas to strictly support UG streams ("judiciary", "clat_ug", "punjab_university_ug"). Purge all PG enums. • Refactor StudentFee schema: completely remove installment arrays. Replace with single totalFee, paymentStatus ("Paid in Full" | "Pending"), paidAt, paymentMode, invoiceNumber. • Build backend fee controller for one-time payment queries and receipts.	backend/src/models/user.model.ts backend/src/models/studentFee.model.ts backend/src/models/course.model.ts backend/src/controllers/fee.controller.ts backend/src/routes/fee.routes.ts	**tsc -b passes. Fee document stores single one-time payment. No installment fields exist.**
Day 2	**Dynamic Onboarding & Cohort Sidebar Routing**	• Overhaul OnboardingPage.tsx: Step 1 presents visual stream selection cards (Judiciary / CLAT UG / Panjab University Law). Step 2 dynamically loads exams for selected stream. • In student-sidebar-items.ts, build getSidebarItemsForStream(). • In StudentSidebar.tsx, render distinct tabs per stream. CLAT UG gets NLU mocks and GK, hiding Mains DAW. PU Law gets CET mocks and Department notes.	frontend/src/pages/auth/OnboardingPage.tsx frontend/src/navigation/student-sidebar-items.ts frontend/src/components/dashboard/StudentSidebar.tsx backend/src/controllers/auth.controller.ts	**CLAT UG student logs in and sees dedicated CLAT sidebar. Judiciary student sees DAW and Bare Acts.**
Day 3	**CLAT Passage Engine & PU Test Series UI**	• Extend mcqQuestion.model.ts with passageText and passageRefCode. • Build POST /api/v1/tests/passage-set. • Overhaul TestExamModal.tsx: Implement split-screen mode for CLAT UG. Left column renders reading passage (scrollable with A-/A+ font resizer); right column renders question 1 of 5. • Configure standard 100-MCQ speed test mode for PU CET.	backend/src/models/mcqQuestion.model.ts backend/src/controllers/testExam.controller.ts frontend/src/components/tests/TestExamModal.tsx frontend/src/services/testEngine.service.ts	**Taking a CLAT test displays passage pinned on left with child questions on right. PU test loads cleanly.**
Day 4	**Cohort-Specific Knowledge Vaults & PYQs**	• Update PyqArchivePage.tsx and admin PyqsPage.tsx with stream filter tabs: All, State Judiciary, CLAT UG Solved Papers (2015-2025), PU 5-Yr & 3-Yr LLB Entrance Archive. • Build Error Notebook (Mistake Vault) in TestResultsPage.tsx: auto-aggregates all questions student missed across tests with explanations and re-quiz trigger.	frontend/src/pages/dashboard/PyqArchivePage.tsx frontend/src/pages/dashboard/tests/TestResultsPage.tsx admin/src/pages/vault/PyqsPage.tsx backend/src/controllers/pyq.controller.ts	**Student reviews only CLAT or PU past papers. Mistake notebook correctly populates wrong MCQs.**

5. Accelerated 7-Day Developer Sprint Plan (Day 5 to Day 7)

Day	Core Focus	Concrete Implementation Tasks & Code Changes	Files Modified	Acceptance Criteria
Day 5	**One-Time Fee Portal & GST Invoice Generator**	• In frontend FeesPage.tsx, replace mock data with live call to /fees/my-status. Display single course fee status: Paid in Full or Payment Pending. • Build generateInvoicePdf.ts using jsPDF: generates professional GST Tax Invoice with RLA logo, student roll no, course name, HSN code, CGST/SGST breakdown, and authorized signature. • Update admin FeesPage.tsx with "Record Full Payment" modal.	frontend/src/pages/dashboard/FeesPage.tsx frontend/src/utils/generateInvoicePdf.ts admin/src/pages/campus/FeesPage.tsx backend/src/controllers/fee.controller.ts	**Clicking "Download GST Invoice" compiles authentic tax PDF. Admin marks fee paid with 1 click.**
Day 6	**Admin Multi-Cohort Switcher & Fast Admissions**	• In AdminLayout.tsx, add master Stream Switcher dropdown ("All Cohorts", "Judiciary", "CLAT UG", "PU Law"). Filters Dashboard KPIs, Roster, Courses, and Tests. • In LeadsPipelinePage.tsx, connect Kanban stages to /leads API. • In LeadsNewEnrollmentPage.tsx, implement 60-second Fast Enrollment: creates student, assigns batch, records one-time payment, and auto-generates roll no. • In TestsQuestionBankPage.tsx, add CSV Question Uploader.	admin/src/components/layout/AdminLayout.tsx admin/src/pages/leads/LeadsPipelinePage.tsx admin/src/pages/leads/LeadsNewEnrollmentPage.tsx admin/src/pages/exams/TestsQuestionBankPage.tsx	**Admin switches cohort filter; data updates instantly. Fast admission creates active student record.**
Day 7	**Printable Digital ID, Gate Scan & Production Launch**	• In StudentIdPage.tsx, implement Download Printable ID Card using html2canvas + jsPDF with academy seal, student photo, cohort, and barcode/QR gate token. • Implement backend POST /api/v1/campus/verify-gate-pass for turnstile scanning. • Run full verification: npm run build in frontend and admin, tsc in backend. • Deliver signed production deployment release.	frontend/src/pages/dashboard/StudentIdPage.tsx backend/src/controllers/campus.controller.ts frontend/ (vite build) admin/ (vite build) backend/ (tsc)	**Zero compile errors. Printable ID PDF downloads cleanly. Full CRM operational in 7 days.**

6. Final Production Acceptance & Scope Compliance Checklist

1. ZERO INSTALLMENTS: Not a single "installment" label, field, or dropdown exists anywhere. Payments are strictly one-time full payments with instant GST invoice compilation.
2. STRICT UG SCOPE: Not a single Post-Graduate (PG) string, course, or PYQ paper exists in the database or user interfaces.

3. DEDICATED SIDEBARS: Judiciary, CLAT UG, and Panjab University students see distinct, customized sidebars matching their entrance pattern.

4. CLAT PASSAGE ENGINE: Split-screen reading passage with child questions works seamlessly on desktop and responsive mobile.

5. ZERO COMPILE ERRORS: All three workspaces (frontend, admin, backend) compile cleanly with npm run build in under 1 second.